

FORTRESS V3

From morning chaos to prioritised action list in 60 seconds.

The edge isn't the data. The edge is the system.

19	3	9	60s
TICKERS IN UNIVERSE	LIVE DATA SOURCES	DASHBOARD PAGES	MORNING WORKFLOW

QuantData MCP	IBKR CP Gateway	yFinance	React 19 + TypeScript
Python FastAPI	Self-Hosted VPS	No SaaS Subscription	

The Problem Every Options Trader Knows

Managing an active options book means juggling five browser tabs, two spreadsheets, a broker terminal, and a chat window — before you've even had coffee.

BEFORE FORTRESS V3

The average morning workflow for an active options trader takes 45–60 minutes: manually checking each position for delta drift, earnings proximity, and roll windows; pulling IV rank data from one tool, dark pool levels from another, and GEX walls from a third; cross-referencing everything against a strategy document stored in a separate file. By the time you have a prioritised action list, the market has already been open for an hour.

PAIN

Fragmented Data

IV rank in one tab. Dark pool floors in another. Greeks from the broker. Earnings dates from a calendar. **No single source of truth.**

PAIN

Manual Cross-Referencing

Every decision requires manually checking the position against the strategy rules. **Discipline breaks down under time pressure.**

PAIN

No Regime Awareness

Entering a new position without knowing whether the macro regime supports it is **flying blind**. Most tools don't surface this at all.

PAIN

Reactive, Not Proactive

Without a systematic screener, you only act on positions that are already in trouble. **The best entries are missed entirely.**

AFTER FORTRESS V3

One dashboard. One morning workflow. One prioritised action list. Fortress V3 aggregates live Greeks from IBKR, institutional flow data from QuantData, and price/IV data from yFinance into a single interface that surfaces exactly what needs attention — and nothing else.

A 4-Layer Decision Engine

Every trade decision flows through four sequential gates. No layer can be skipped. Each one narrows the field and raises the quality of the final action.

LAYER 01 Regime Gate Macro score from GEX, dark pool & net drift. No new entries when score $\leq$ 0.	→	LAYER 02 Market Intel Per-ticker GEX walls, DP floors, IV rank. Identifies the strongest setups.	→	LAYER 03 Position Eval Live Greeks, delta drift, DTE triage, concentration limits. Surfaces what needs action.	→	LAYER 04 Priority Orders URGENT STOP-LOSS → ROLL → CLOSE → NEW_ENTRY. Ranked by urgency.
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The 9 Dashboard Pages

Each page serves a specific function in the workflow. No page is decorative.

PAGE	FUNCTION	KEY DATA
Dashboard	Morning brief, regime score, P&L	Net Liq • Regime Score • Priority Orders
Candidates	IV screener — 19-ticker universe	IV Rank • IV/HV Spread • Signal
Market Intel	Institutional flow per ticker	GEX Walls • DP Floors • Net Drift
Analysis	Chart with position overlays	Price • Greeks • Chart Annotations
Portfolio	Per-leg Greeks and alerts	Delta • Theta • DTE • Concentration
Performance	P&L history and attribution	Realised P&L • Win Rate • Drawdown
Earnings	Earnings calendar and blackouts	Earnings Date • Blackout Window
Positions	Full book with IBKR sync	All Legs • Cost Basis • MV
Settings	Strategy parameters, credentials	9 Tunable Parameters • 5 Presets

Three Live Data Sources, Unified

Fortress V3 aggregates institutional-grade data from three independent sources into a single normalised data layer. No manual imports. No stale spreadsheets.

QD

QuantData MCP + REST

Institutional options flow data via the QuantData MCP server and widget-UUID REST endpoints. Provides:

- GEX Call Wall and Put Wall per ticker
- Dark Pool Floor and Ceiling levels
- Net Drift (directional bias score)
- IV Rank and IV/HV spread
- Order flow consolidated data

Credentials managed via Settings → QuantData Credentials. No manual config file edits required.

IBKR

Interactive Brokers CP Gateway

Live broker data via the CP Gateway (voyz/ibeam Docker container). Provides:

- Live Greeks (delta, theta, vega, gamma) per leg
- Real-time position market values
- Account metrics: Net Liq, Excess Liq, Available Funds
- Option chain snapshots for strike selection
- SPY hedge market value tracking

Fallback: Black-Scholes Greeks from yFinance when CP Gateway is unavailable.

YF

yFinance + Price Data

Price and volatility data for all 19 universe tickers. Provides:

- Current price and daily OHLCV
- 200-day SMA (stop-loss signal input)
- Historical volatility (HV) for IV/HV spread
- Earnings date lookup
- Post-earnings IV crush measurement

Used as the always-available data layer when institutional sources are unavailable.

TV

TradingView Integration

Chart analysis and alerts via TradingView. Provides:

- Clean Decision Chart for strategic decisions
- Signal/Timing Chart for tactical entries
- Price alerts delivered to dashboard
- 200-SMA breach notifications

The Analysis page in Fortress V3 overlays open positions directly onto the price chart — short call/put strikes, LEAP entry level, GEX walls, and earnings markers all visible without leaving the dashboard.

QUANTDATA MCP — TECHNICAL NOTE

Fortress V3 uses the QuantData MCP server with widget-UUID-based REST endpoints (`/api/options/exposure/strike/{uuid}`, `/api/equities/dark-pool/levels/{uuid}`, `/api/options/net-drift/{uuid}`). The deprecated `tool/OPTIONS_*` tool IDs are not used. All three data-consuming scripts (`market_intelligence.py`, `chart.py`, `iv_crush_scanner.py`) have been updated to use the correct endpoints.

Key Features

Every feature exists to reduce cognitive load and enforce the trading strategy. Nothing is decorative.

BRIEF

Morning Brief

One-click prioritised action list generated from live data. **URGENT STOP-LOSS → ROLL → CLOSE → NEW_ENTRY** — ranked by urgency. From zero to action list in 60 seconds.

SCAN

IV Rank Screener

Full 19-ticker universe screened against IV Rank, IV/HV spread, and regime score. **Actionable signals** (STRONG_SELL/SELL/WATCH) at top; monitoring rows below. Always shows the full universe even when no signals are active.

INTEL

Market Intelligence

Per-ticker institutional flow dashboard. **GEX walls, dark pool levels, net drift, directional bias score.** Sort by score, bias, or alphabetical. Per-card refresh. Hover tooltips explain every metric.

RISK

Position Risk Monitor

Live Greeks per leg with **delta drift alerts** (amber at 0.30–0.35, red above 0.35), DTE triage badges, concentration percentage, and roll prompts. All thresholds configurable in Settings.

GATE

Pre-Trade Gates

Hard gates enforced before every new entry: **earnings blackout, concentration limit, regime score, margin floor, SPY hedge coverage.** Visual warnings for soft limits; blocks for hard limits.

CFG

Schema-Driven Settings

9 tunable parameters (delta threshold, IV rank floor, regime gate, margin floors, concentration limits) and **5 trader presets** — all configurable in the UI without a code deploy.

9

TUNABLE PARAMETERS

5

TRADER PRESETS

4

DECISION LAYERS

Production-Grade Stack

Self-hosted on a dedicated VPS. No cloud lock-in. No SaaS subscription beyond data providers. Full control over data, logic, and infrastructure.

FRONTEND

- **React 19** + TypeScript + Tailwind CSS 4
- **tRPC 11** + React Query (type-safe end-to-end)
- **Dark terminal UI** — phosphor green on near-black
- **60-second polling** — live data without WebSockets
- **shadcn/ui** component library

BACKEND

- **Python 3.11** + FastAPI + uvicorn
- **Bearer token** authentication on all endpoints
- **systemd** service — always-on, auto-restart
- **MySQL/TiDB** for positions and history
- **S3** for reports and exports

DATA LAYER

- **QuantData MCP** — institutional flow (GEX, DP, drift)
- **IBKR CP Gateway** — live Greeks and account data
- **yFinance** — price, SMA, HV, earnings
- **8 workflow scripts** — pre-market to EOD

INFRASTRUCTURE

- **Dedicated VPS** — 76.13.138.194
- **nginx** reverse proxy
- **GitHub** CI/CD pipeline
- **Manus** deployment layer

REST API — 9 Endpoints

All endpoints are bearer-token authenticated. Fully programmable — usable from scripts, bots, or external tools.

FORTRESS V3 · OPTIONS TRADING COMMAND CENTRE

FORTRESS V3

Stop managing tabs.
Start managing risk.

"The edge isn't the data. The edge is the system."

QuantData MCP

IBKR CP Gateway

yFinance

React 19 + Python

Self-Hosted · No Lock-In

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